

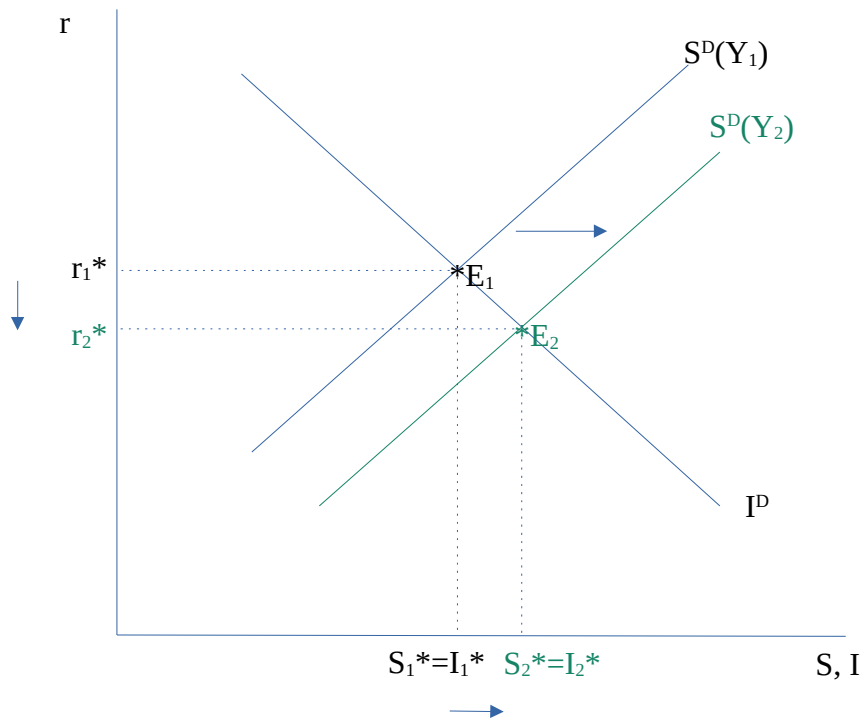
University of Waterloo
Department of Economics
ECON 202-002&-003 W2025
Assignment # 2
Solutions.
parts in orange not required

1. Explain and graph the impact on the saving demand curve, investment demand curve, equilibrium r^* , S , and I , of the following changes:

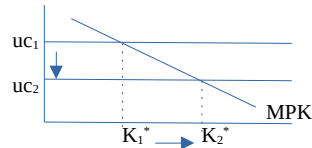
a) Current Y rises. (5)

	Impact	Explanation
r^* :	↓	$S^D = Y - C^D - G$, so $\uparrow Y \rightarrow S^D \uparrow$ (shift right)
S^* :	↑	
I^* :	↑	$I^D = K^* - K_t + dK_t$, where K^* determined where $UC = MPK^f$, none of which depend on Y With S^D increasing then $r^* \downarrow$ (@ r_1^* , demand for funds (I^D) < supply of funds (S^D) → downward pressure on r until $S=I$) @ new r_2^* , S^* is higher and I^* is higher
I^D curve:	No change	
S^D curve:	↑ (shift right)	

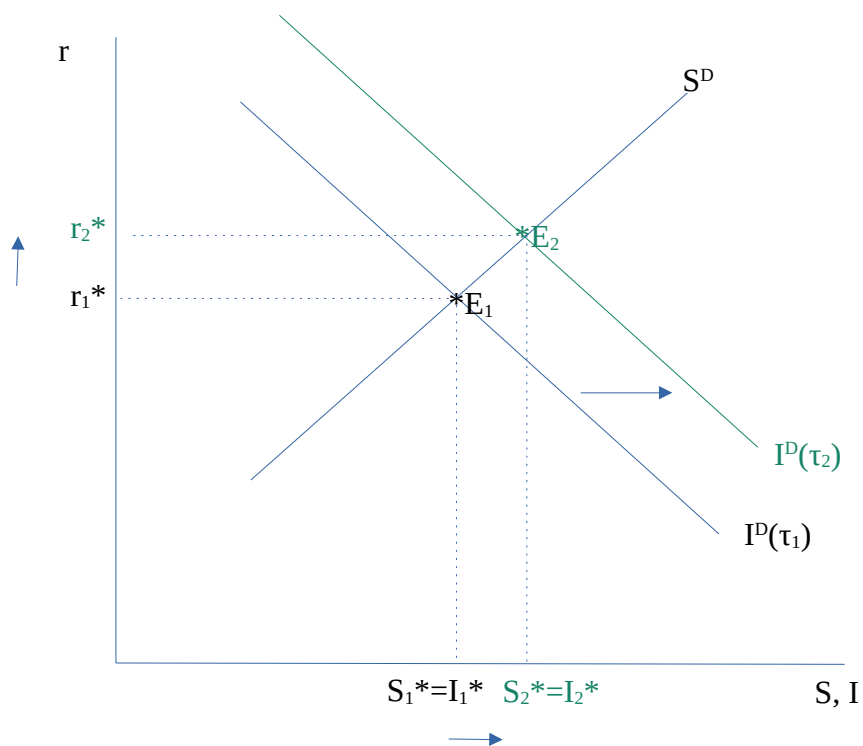
Graph



b) A new government is elected and cuts the effective tax rate on capital. (5)

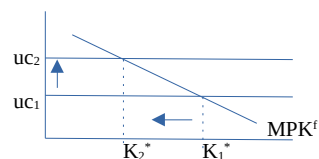
	Impact	Explanation
r^* :	↑	A tax on capital can be modelled as $(1-\tau)MPK^f$ or $UC/(1-\tau)$. I choose the latter for simplicity. K^* is determined where $UC/(1-\tau)=MPK^f$ This graph not required 
S^* :	↑	
I^* :	↑	
I^D curve:	↑ Shift right	If $\tau \downarrow$ then the user cost of capital $\downarrow \rightarrow$ profit maximizing $K^* \uparrow$ $K^* \uparrow \rightarrow I^D \uparrow$ (shifts right) ($I^D = K^* - K_t + dK_t$, so $\uparrow K^* \rightarrow I^D \uparrow$)
S^D curve:	No change	With I^D increasing then r^* will rise. (@ r_1^*, demand for funds (I^D) > supply of funds (S^D) $\rightarrow$ upward pressure on r until $S=I$) @ new r_2^*, S^* is higher and I^* is higher

Graph



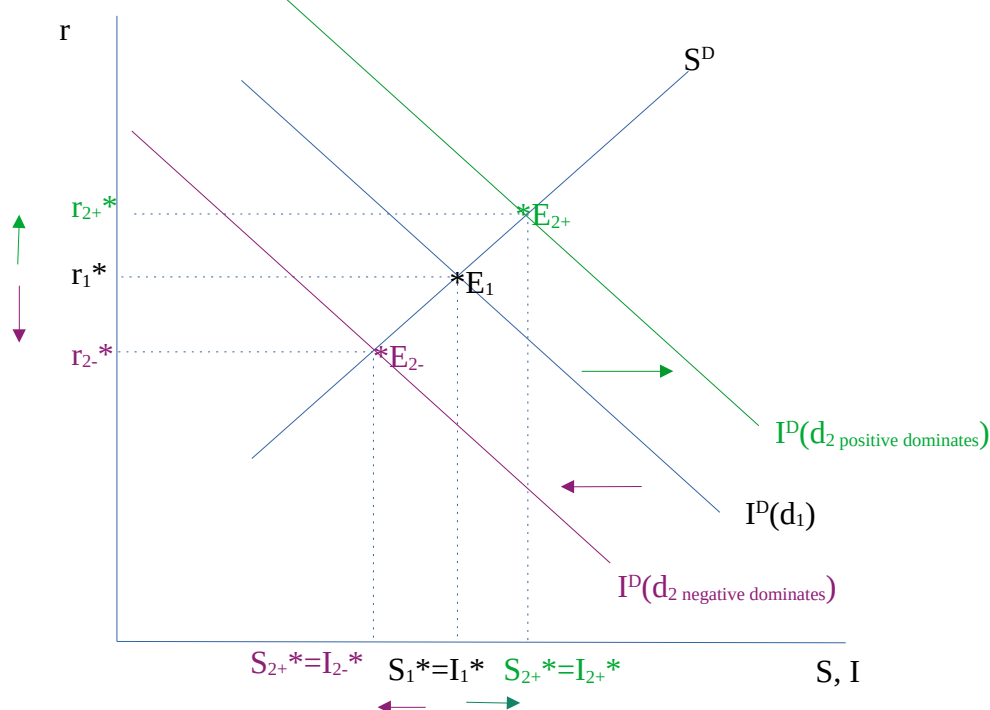
c) The depreciation rate of capital rises. (6)

	Impact	Explanation
r*:	ambiguous	K^* is determined where $UC=MPK^f$ $UC = (r+d)*P_K$ $\uparrow d \rightarrow UC \uparrow \rightarrow K^* \downarrow$
S*:	ambiguous	
I*:	ambiguous	
I^D curve:	ambiguous	
S^D curve:	No change	



This graph not required

Graph



2. Consider how our fractional reserve banking system influences money supply via the multiple expansions of loans and deposits.

a) What are the implications of a lower currency deposit ratio on the money supply? Explain. (2)

Implications:	Explanation:
M ↑	Less money in pocket rather than in the bank, means an increase in the bank's expansion of the money supply (via the multiple expansion of loans and deposits). Algebraically, the money multiplier = $(cu+1)/(cu+res)$, and since $res < 1$ ↓ cu → denominator ↓ by more than numerator, so the entire ratio ↑ M = money multiplier * BASE, so ↓ cu → money multiplier ↑ → M ↑

b) From a bank's perspective, list one advantage and one disadvantage of a low reserve deposit ratio.(2)

Advantage:	Low res → more lending → more profits for bank
Disadvantage:	Low res → higher risk of not having funds to pay when folks come to withdraw deposits (although this is a pretty low risk in Canada given all of the safety regulations and of course the deposit insurance, so banks runs are unlikely.)

c) What would happen to money supply if the reserve deposit ratio were increased to 1 (100%)? Explain. (3)

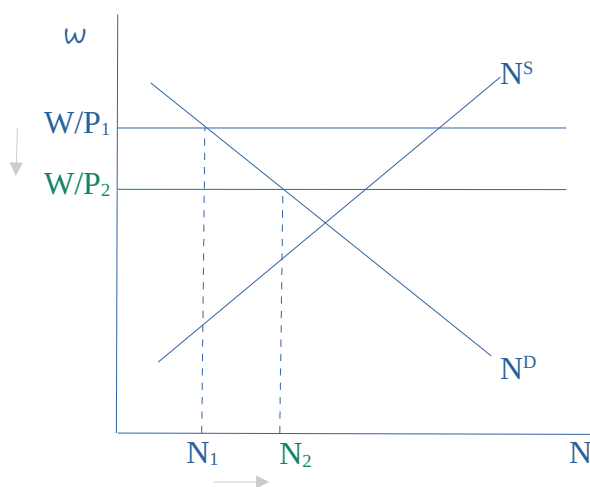
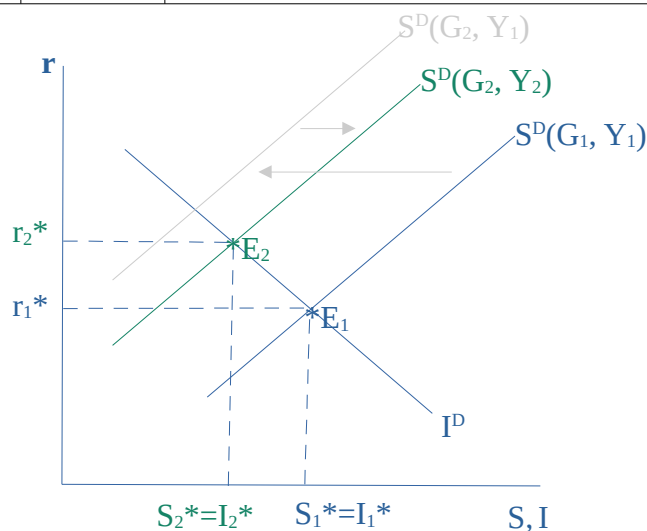
Impact on Money supply:	M ↓
Explanation:	If res increased to $res=1$ then no fraction of deposits would be lent out anymore, so the banks would not be expanding money supply anymore, and therefore money supply falls. Algebraically, the money multiplier = $(cu+1)/(cu+res)$, and since initially $res < 1$, then the multiplier would be larger than 1 and the money supply would have started out larger than the base. M = money multiplier * BASE If res increases to 1, then the numerator and denominator are identical, so the ratio = 1. M = money multiplier * BASE → M = 1 * BASE, that is M = BASE Therefore, when res increases to 1, then M ↓ and now equals the BASE.

3. Answer the following questions:

a) Graph and explain the impact of increased government spending on S^* , I^* , r^* , Y , P , N & w in sticky wage framework. (8)

	Impact	Explanation
r^* :	rise	$\uparrow G \rightarrow \downarrow S^D = Y - C - G \uparrow \rightarrow S^D \text{ shifts left} \rightarrow r^* \uparrow, S^* \downarrow, I^* \downarrow$ $AD: \uparrow Y = C^D + I^D + G \uparrow \rightarrow AD > AS \rightarrow P \uparrow$
S^* :	fall	
I^* :	fall	
Y :	rise	
P :	rise	$\uparrow Y \rightarrow S^D \uparrow$ (a bit – slight shift right but doesn't override first shift) $\uparrow P \rightarrow \frac{W}{P} \downarrow \rightarrow N \uparrow$
N :	rise	(As discussed in class, in the sticky wage framework, W is sticky downward --it doesn't decrease--, so a rise in price will reduce real wage (W/P), but if that rise in price were very large and W/P fell below w^* , W could adjust upward. For the purposes of this assignment, I will accept ambiguous effect on N as an answer since I did not recall this element when first introducing the model)
w :	fall	

Graphs



b) Did the Canadian economy experience any of these effects as Government spending increased partway through the covid pandemic? Explain. (2)

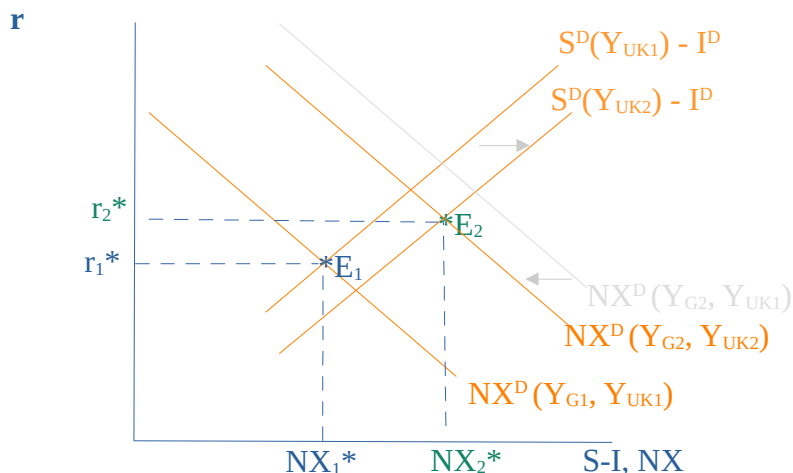
Answer:	Yes.
Explanation:	Prices rose (inflation) Of course, prices rose not just because of expansionary fiscal and monetary policy (including quantitative easing), but also because supply fell. (Some suggest that the expansionary policies had little impact relative to the supply effect). Note that N fell, w rose, S rose on average because C fell. Interest rose over COVID, but inflation rose more, so real interest rates fell. Therefore, with the exception of Prices, pretty much everything else went in the opposite direction of this example (which is likely due to the massive supply effect). Note, Y dropped steeply, but it did rebound, and this rebound may have been helped by expansionary policy. So an increase in Y would be an acceptable answer too.

4. Suppose Germany uses an expansionary monetary policy resulting in higher Y in Germany. Assume the Y effect dominates changes in NX in Germany and abroad and answer the following questions.

a) How will Germany's monetary policy impact the economy of their trading partner the UK? Specifically, how will the UK's Y , r , and NX change? (4)

	Impact	Explanation
NX_{UK}:	rise	$\uparrow Y_G \rightarrow$ Germany imports more UK goods $\rightarrow \text{Exp}_{UK} \uparrow$ $\rightarrow NX_{UK} \uparrow$ (shift right) $\rightarrow r^* \uparrow$
r_{UK}:	rise	$AD: \uparrow Y_{UK} = C^D + I^D + G + NX^D \uparrow$ $\uparrow Y_{UK} \rightarrow S^D \uparrow \rightarrow S^D - I^D$ shifts right (a bit, doesn't override initial impact) $\rightarrow NX^D \downarrow$ (shifts left a bit, doesn't override initial impact)
Y_{UK}:	rise	(Technically, the effects could be ambiguous but you were asked to assume that the Y_G effect dominates)

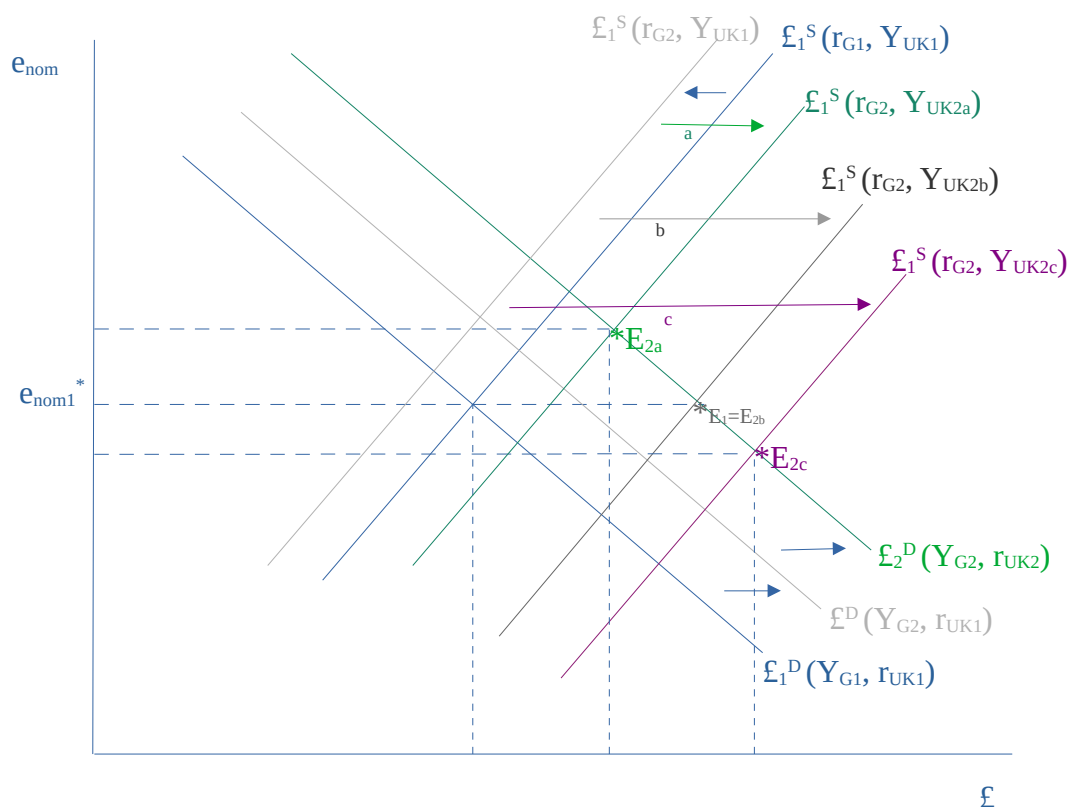
Extra space if needed (you are not required to put anything here, but I have added a graph to help illustrate)



b) What happens to the UK exchange rate? Use a diagram for UK's exchange market to help you explain what happens to e_{uk} . (5)

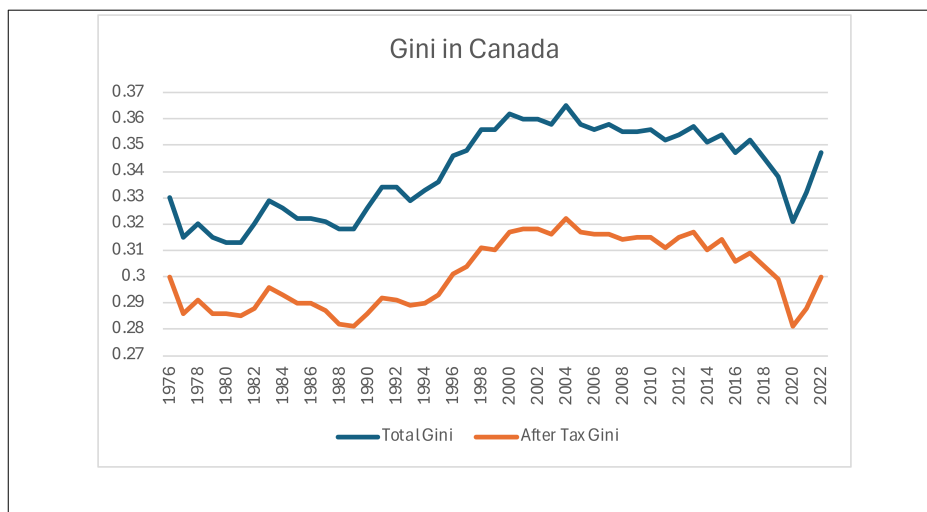
	Impact	Explanation
e_{UK} :	ambiguous	$\uparrow Y_G \rightarrow NX_{UK} \uparrow$ (due to higher exports) $\rightarrow \pounds^D \uparrow \rightarrow e_{nomUK} \uparrow$ $\uparrow Y_{UK} \rightarrow NX_{UK} \downarrow$ (due to higher imports) $\rightarrow \pounds^S \uparrow \rightarrow e_{nomUK} \downarrow$ $\uparrow r_{UK} \rightarrow \text{invest into UK} \uparrow \rightarrow \pounds^D \uparrow \rightarrow e_{nomUK} \uparrow$ $\uparrow M_G \rightarrow r_G \downarrow$ $\downarrow r_G \rightarrow \downarrow \text{investment into Germany} \rightarrow \pounds^S \downarrow \rightarrow e_{nomUK} \uparrow$ so e may rise or fall, depending on which effect dominates. All shifts shown below. The effect of $\&$ are both to increase demand. The effects of $\&$ move in opposite directions, so it will depend on the relative magnitude of the Y_{UK} vs r_G effects. I demonstrate by showing three potential relative shifts (a, b, c - green, grey, purple). If the Y_{UK} Effect is relatively smaller (green), then $e_{nomUK} \uparrow$, if the Y effect is relatively larger (purple), then $e_{nomUK} \downarrow$. It is also possible for the two effects to perfectly balance (grey) in which case e_{nomUK} won't change. Although typically, it is unlikely that the Y_{UK} effect would override the others.

Graph of UK exchange market

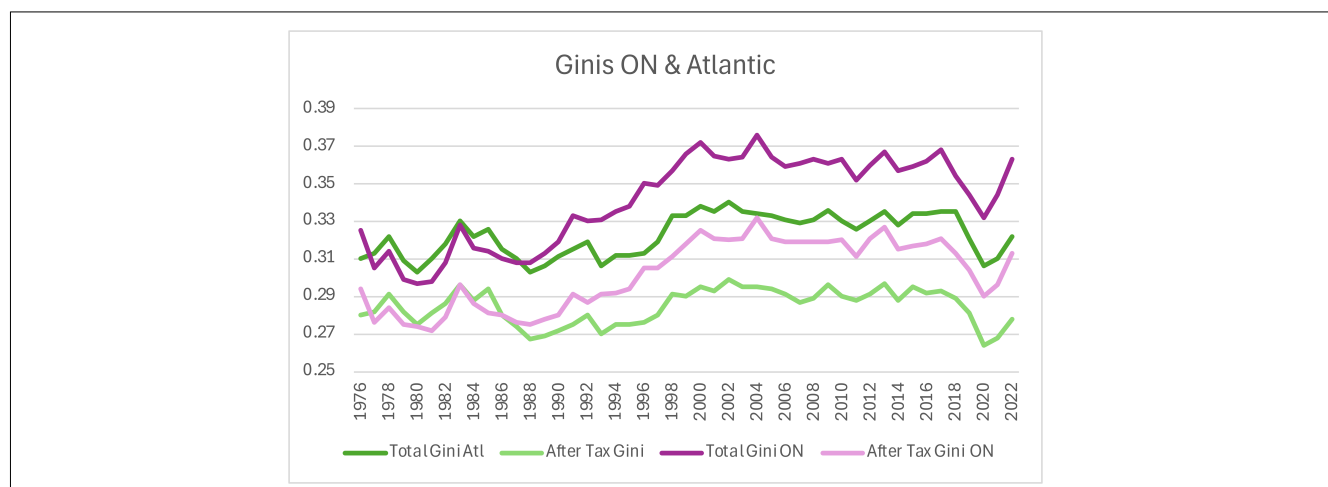


5. Access Statistics Canada's table 11-10-0134-01 and download the Gini coefficients for total income and after tax income for Canada, Ontario, and the Atlantic Provinces from the earliest to the most current date.

a) Graph these two Gini measures for Canada across time. What do you observe about the Gini for total income vs after-tax income? (2)



b) Compare Gini trends for Ontario and the Atlantic Provinces. What do you observe? (4)



Observations:

Note: Purple/Pink are Ontario, Green are Atlantic. Lighter colours are after-tax. You only need to show one set of series for each region. But showing both for both is fine too.

Potential observations you could make:

*Whether you use Total or after-tax Ginis, the Ginis between Ontario and Atlantic provinces used to be similar (in the 1970s). Since then, Ginis in Ontario have risen higher than the Atlantic provinces. That is, inequality has become larger in Ontario than in the Atlantic provinces.

*The Ontario and Atlantic Ginis tend to spike and dip at roughly the same times. E.g. at the start of Covid, the Ginis fell dramatically in 2020 for both regions, and then increased rapidly for both regions thereafter (more so for Ontario).

*For both Ontario and Atlantic, the After Tax Gini is lower than the before tax (Total) Gini. That suggests both provinces tax systems redistribute from the higher to the lower earners.

*For both regions, using either measure, the Gini rose substantially during the 1990s, and has remained relatively high ever since. (Higher inequality).

*About the time of 2008 financial crisis, Ginis rose then dipped down below their initial value by 2011. More so for the Atlantic than Ontario.